

Nishan Rai

M.S. Computer Science · New Mexico State University

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Education

New Mexico State University (NMSU)

Jan 2025 – Present

Master of Science in Computer Science — Las Cruces, NM

GPA: 4.0/4.0 — GTA & Graduate Assistantship

Institute of Engineering, Tribhuvan University

Nov 2018 – Aug 2023

B.E. in Electronics, Communications and Information Engineering — Nepal

- Graduated with Distinction — 80.41% — Ranked top of cohort, 2nd–8th semester

Research Experience

Graduate Researcher — New Mexico State University

Aug 2025 – Present

Supervisor: Dr. Tuan Le — Dept. of Computer Science

- Developing a multi-view mathematical formula retrieval framework that combines graph neural networks (capturing structural formula representation) with transformer-based contextual embeddings (capturing semantic context).
- Evaluating the framework on the NTCIR-12 Math benchmark using standard TREC evaluation metrics.

Preprints

1. **Nishan Rai**, S. Khatri, D. Risal.

Explainable AI Technique in Lung Cancer Detection Using Convolutional Neural Networks.

arXiv preprint arXiv:2508.10196, 2025.

2. **Nishan Rai**, P. R. Dahal.

A Unified Attention U-Net Framework for Cross-Modality Tumor Segmentation in MRI and CT.

arXiv preprint arXiv:2601.06187, 2026.

Research Projects

Explainable AI for Lung Cancer Detection

- Developed and benchmarked CNN architectures (DenseNet121, ResNet152, VGG19, custom CNN) for lung cancer detection from CT scan images.
- Applied SHAP-based post-hoc explainability to identify discriminative CT scan regions, improving model trustworthiness for clinical users.
- Awarded Best Poster Presenter at the 16th NeSA International Conference (2025).

Cross-Modality Tumor Segmentation using Attention U-Net

- Developed an Attention U-Net framework for tumor segmentation transferable across MRI (BraTS) and CT (LIDC-IDRI) modalities.
- Designed custom hybrid loss functions (Dice + Focal) and advanced augmentation strategies to address class imbalance in medical imaging datasets.
- Achieved competitive Dice scores; results written up as arXiv preprint (arXiv:2601.06187).

Teaching Experience

Teaching Assistant — Computer Science Department, NMSU

Jan 2025 – Dec 2025

- **Spring 2025** — CSCI 2230 / Assembly Language and Machine Organization: Assisted lab sessions, graded assignments with detailed feedback, and provided individual support during lab hours.

- **Fall 2025** — CSCI 2210 / Object-Oriented Programming using C++: Assisted lab sessions, graded assignments, and supported students through responsive communication and office hours.

Professional Experience

Web & CRM Systems Graduate Assistant (GA) — Graduate School, NMSU *Feb 2025 – Present*
Spring 2025, Fall 2025, Spring 2026 (part-time); Summer 2025, Summer 2026 (full-time)

- Manage and maintain Graduate School web infrastructure using Cascade CMS, ensuring accurate and timely content updates.
- Support digital workflows and user access management in Slate CRM, optimizing operational efficiency for admissions processes.
- Troubleshoot and resolve system-level issues; provide ongoing technical support across institutional platforms.

Business Intelligence Executive — Foreveryng, Nepal *May 2023 – Dec 2024*

- Built data-driven reporting and analytics workflows in Python and SQL, consolidating multi-source data for executive decision-making.
- Automated recurring forecasting pipelines and delivered interactive dashboards in Tableau and Power BI for KPI tracking.

Technical Skills

Programming: Python, C/C++, SQL, MATLAB

ML / DL: PyTorch, TensorFlow, HuggingFace Transformers, PyTorch Geometric, Scikit-learn

Tools: Git, Linux, Jupyter, L^AT_EX

Awards & Honors

- **Best Poster Presenter**, 16th NeSA International Conference, 2025
Presentation: *Enhancing Lung Cancer Diagnosis with Explainable AI and Deep Learning*
- Academic Distinction — Ranked top of cohort across 2nd–8th semesters, B.E. program, Tribhuvan University.